

Assignment 2

Causal Inference

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1 Descriptive Analysis

1.1 Descriptive data

The total sample consist in 158 units, divided equally by treatment/control unit (79). The sample mean reveal an initial unbalance: The treatment units

are on average younger, $T = 64.65, C = 68.4$, less likely to live in rural areas with a significant imbalance, $T = 36.7\%, C = 67.1\%$ and less like to be male $T = 68.4\%, C = 81\%$ (see Figure 2), even though male units are the majority in both groups.

1.2 Balance Analysis

All covariates are unbalanced ($|\Delta| > 0.1$), with particularly pronounced imbalance in the **rural** variable. The age covariate shows greater dispersion in the treated group. The distributions differ significantly between groups, making adjustment methods necessary for a reliable causal effect estimation.¹

2 Stratification Based on the discrete covariates

2.1 Costruction and strata analysis

Are defined 12 strata based on the combination of covariates. One strata ($[25 - 60)$, rural, female) shows a breach of the overlap, containing only 2 treatment units and no one of control. There are also two strata with either one treatment or one control unit, but non-zero for both. These are ($[25 - 60)$, urban, female) and (> 75 , rural, female). Nine strata satisfy the condition of having at least two units in the treated group and control group. Unit proportions shown in Table 2.

2.2 Stratified analysis

Drop the strata with a inadequate support, the estimate of the average treatment effect (ATE) was calculated as a weighted average of the effects within the strata. The estimated effect is $\hat{\tau} \approx 0.103$ with a variance of 0.073. The corresponding 95% confidence interval is $[-0.074, 0.280]$. Since the interval includes zero, there is no statistical evidence of a significant causal effect of hospital volume on one-year survival.

3 Outcome Regression

To address selection bias and effect heterogeneity, we implemented the (Lin) estimator using OLS with full interactions between treatment Z_i and centered covariates $\tilde{\mathbf{X}}_i$ ($Y_i \sim Z_i * \tilde{\mathbf{X}}_i$). Centering ensures the treatment coefficient directly identifies the Average Causal Effect (ACE), estimated at $\hat{\tau}_{Lin} = 0.033$ with HC3 robust standard errors of 0.087 ($p = 0.70$). The result confirms a statistically null causal effect; the adjustment shows that initial survival disparities were driven by covariate imbalances rather than hospital size, as the effect remains indistinguishable from zero after controlling for patient profiles.

4 Estimate Propensity score

Treatment assignment probabilities were estimated via logistic regression on the observed covariates. The resulting Propensity Scores (PS) average 0.57 for the treated and 0.43 for the controls. While the common support ranges from 0.21 to 0.88, the distributions exhibit significant imbalance in the tails. This partial overlap confirms that adjustment or trimming is necessary to ensure comparability and satisfy the overlap assumption before estimating causal effects.

4.1 Discarding Control Units

We computed propensity scores minimum and maximum for Treated units; control units falling out of this range were discarded. Table 3 shows the range. In this phase we discarded those Control units *not comparable* wrt Treatment units.

4.2 Re-estimation of Propensity Scores

Now, after removing those 4 units, we re-computed the propensity score...

4.3 Distribution Comparison and Balance Evaluation

The propensity scores for Treated range from 0.2348 to 0.8661 with mean 0.5730; Control units show a mean 0.4497 and range from 0.2398 to 0.8661. Fig. 7 shows the overlap on trimmed data. A quantitative evaluation of the Trimming procedure is shown by the balance measure prior and posterior to the trimming $\Delta_{ps,initial} = 0.8188756$ and $\Delta_{ps,trimmed} = 0.7525317$

5 Propensity Score Subclassification

5.1 Subclassification

Working on the trimmed data from the previous section we stratified the sample based on some quantiles (0.25, 0.50, 0.70, 0.85) of the estimated propensity scores. In table 4 we report the number of treated and control units in each strata.

5.2 Covariate Balance Analysis

Table 5 and Figure 8 demonstrate improved covariate balance post-stratification. Table 6 compares the estimates of pseudo-average causal effect of the treatment on each covariate, $\tau_X^{(l)}$, pre and post stratification. While values for rural and male are near zero, age shows a slightly higher absolute value. However, Figure 9 shows that all 95% CIs for these pseudo-outcomes include zero, suggesting good balance.

Results of an ANOVA F-Test for each covariate add further evidence of this (table 8).

5.3 ATE Estimation

Treating the subclassified sample as an SRE yields an ATE estimate of 0.0402 ($SE = 0.0866$; 95% CI: $[-0.1294, 0.2099]$). Since the interval includes zero, we find no evidence of a non-zero average treatment effect on patient survival. Qualitatively similar results were obtained using an adjusted ATE estimate: applying Lin’s estimator (adjusting for age, HC2 as variance estimate) within each stratum and aggregating, we calculate a point estimate of 0.0589 ($SE = 0.0891$).

6 Matching

6.1 Match SubSample

To estimate the ATT, we implemented a 1:1 Mahalanobis distance matching procedure with replacement and exact match on male and rural. This means that a treated unit is paired with a control unit that shares the same sex and geographic values.

6.2 Balance before and after

Table 9 shows the p-value on single variables, the table suggests that the matching procedure introduced a not-negligible balance!

6.3 Causal Effect Estimation

We estimated the Average Treatment Effect on the Treated (ATT) using the Bias-Adjusted estimator. We experienced the following results:

- **Estimated ATT ($\hat{\tau}_{ATT}$):** 0.1210
- **Sample Variance** 0.0083
- **95% Confidence Interval:** $[-0.0574, 0.2994]$

Since the 95% confidence interval contains zero, we are not allowed to reject the null hypothesis of no treatment effect. Meaning that the treatment effect (in the estimated average) on the treated population is not significant with the conventional 5% significance level.

Table 1: Summary table

Covariate	Delta	Gamma
Age	-0.306	0.151
Rural	-0.6434	None
Male	-0.292	None

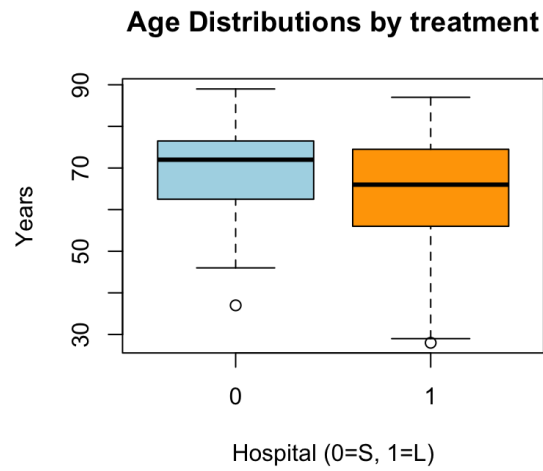
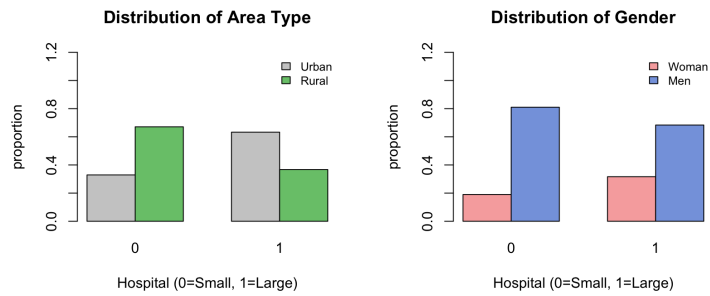


Figure 1: BoxPlot of age distribution from treatment and control groups



(a) BarPlot of Rural Area Distribution of treatments and control units (b) BarPlot of Gender distribution of treatments and control units

Figure 2

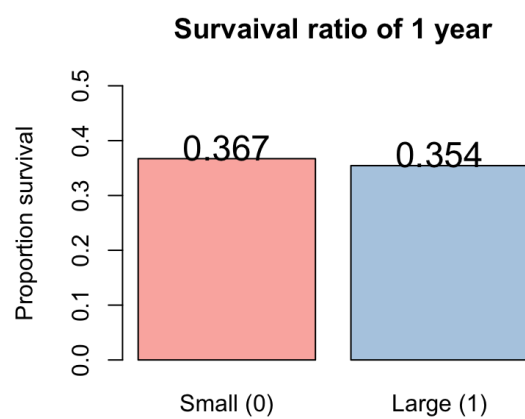
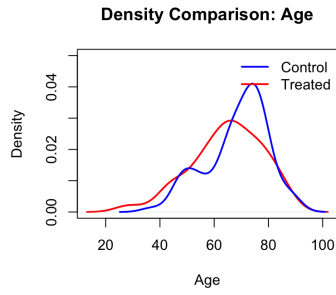


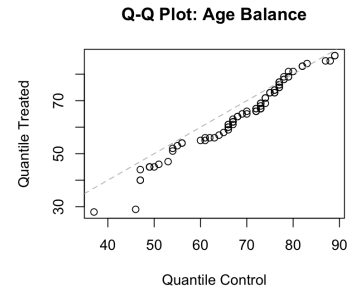
Figure 3: Survival ratio of 1 years

Table 2: SRE Table

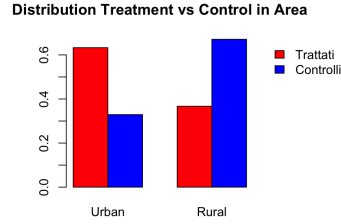
Strata	Number of Strata	Proportion
Non-overlapping strata	1	1.27%
One c or t Strata	2	5.7%
Valid Strata	9	93%



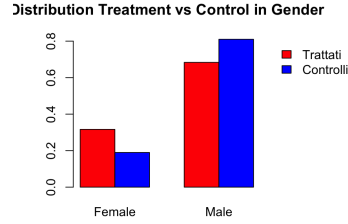
(a) Density Distributions



(b) Q-Q Plot of Age



(a) Treatment vs Control units in Area



(b) Treatment vs Control units in Gender

Table 3: Propensity Scores Minimum and Maximum for Treatment units

PS Minimum	PS Maximum
0.2136436	0.8805467

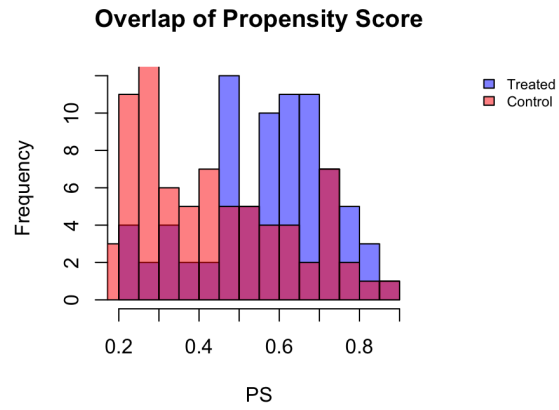


Figure 6: Overlap of Propensity score

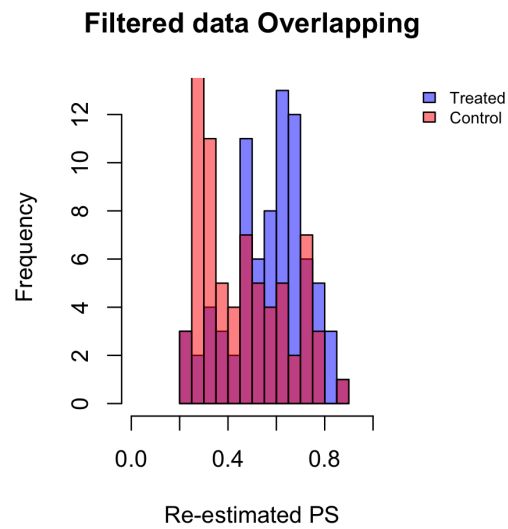


Figure 7: Overlap of Propensity score on Trimmed Data

Table 4: Control and Treated per strata after PS-based stratification

Stratum	Control	Treatment	Total
1	30	9	39
2	20	19	39
3	11	19	30
4	5	18	23
5	9	14	23

Table 5: Control and Treated per strata after PS-based stratification

Statistics	Age	Rural	Male
Δ_{ct}	-0.230	-0.594	-0.267
$\Delta_{ct}^{str_1}$	-0.042	0.000	0.000
$\Delta_{ct}^{str_2}$	0.181	-0.338	0.072
$\Delta_{ct}^{str_3}$	-0.408	0.357	-0.274
$\Delta_{ct}^{str_4}$	0.223	0.115	-0.178
$\Delta_{ct}^{str_5}$	-0.026	0.000	-0.200
$\Delta_{ct}^{str_{all}}$	-0.015	0.001	-0.091

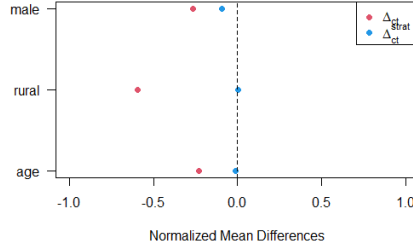


Figure 8: Normalized Mean Differences pre and post stratification

Table 6: Comparison between pre and post stratification $\tau_x^{(l)}$

Covariate	$(\bar{X}_1 - \bar{X}_0)$	$\tau_{x,1}^{(l)}$	$\tau_{x,2}^{(l)}$	$\tau_{x,3}^{(l)}$	$\tau_{x,4}^{(l)}$	$\tau_{x,5}^{(l)}$	$\tau_x^{(l)}$
Age	-2.77	-0.51	2.18	-4.91	2.69	-0.31	-0.18
Rural	-0.29	0.00	-0.16	0.17	0.06	0.00	0.00
Male	-0.12	0.00	0.03	-0.12	-0.08	-0.09	-0.04

Table 7: Variances of table 6 quantities

Covariate	$(V_{\bar{X}_1 - \bar{X}_0})$	$V_{\tau_{x,1}^{(l)}}$	$V_{\tau_{x,2}^{(l)}}$	$V_{\tau_{x,3}^{(l)}}$	$V_{\tau_{x,4}^{(l)}}$	$V_{\tau_{x,5}^{(l)}}$	$V_{\tau_x^{(l)}}$
Age	1.95	2.08	4.20	3.27	6.90	6.91	1.98
Rural	-0.08	0.00	0.12	0.115	0.11	0.00	0.05
Male	-0.07	0.00	0.16	0.14	0.23	0.22	-0.07

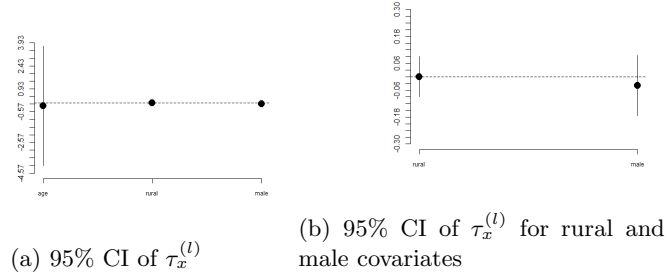


Figure 9

Table 8: P-Value resulting from the ANOVA F-test

Covariate	P-value
Age	0.867
Rural	0.289
Male	0.955

Table 9: Before and After Matching Balance

Covariate	Before Matching	After Matching
Age	0.0563	0.3367
Rural	0.0001	1.0000
Male	0.0681	1.0000